Decentralized Exchanges (DEX) have emerged recently with Uniswap as the leading automated market maker (AMM) by volume. An automated market maker (AMM) is a protocol which permits the execution of buy and sell orders on a blockchain. These protocols play a similar role as limit order books in traditional finance; users deposit their tokens into a liquidity pool, where they can be used by other actors for trading activities, enabling decentralized exchanges. A liquidity provider (LP) is an actor who transfers tokens into liquidity pools and is rewarded by the fees paid by traders who use that pool to swap tokens.

Uniswap v3 is a constant product market maker, meaning that it maintains invariant function $\mathcal{I}(x, y) = x \cdot y = L^2$ where L denotes the liquidity and x and y are virtual reserves of tokens $\mathcal{X}$ and $\mathcal{Y}$, respectively. A virtual reserve for a token $\mathcal{X}$ is defined as $x = x_{\text{offset}} + x_r$ where x_r is the real reserves, and x_{offset} are some other quantities. Liquidity providers wishing to provide liquidity must specify a price range over which their tokens will be used for trading activities (as long as the current price resides in that range). A square-root price range $R := [\pi_l, \pi_u)$ where $\pi_l = \sqrt{p_l} = \sqrt{\frac{x_l}{y_l}}$ and $\pi_u = \sqrt{p_u} = \sqrt{\frac{x_u}{y_u}}$ must be specified in ticks τ. The constant-product rule applies here $x_u \cdot y_u = x_l \cdot y_l = L^2$, which corresponds to a point on the price space (see fig. 1) Specifically, price $p(\tau) = \left(\beta^{(p)}\right)^{\tau}$ where $\beta^{(p)} = 1.0001$ and τ is a multiple of $\delta_\pi \in \mathbb{N}$. A *unitary* price range is the range spanned by two consecutive ticks $p_l(\delta_\pi \cdot i)$ and $p_u(\delta_\pi \cdot (i+1))$.

Consider a square-root unitary price $R = [\pi_l, \pi_u) = R_1 \cup \ldots \cup R_n$, where $R_i = [\pi_{l_i}, \pi_{u_i})$ already contains liquidity L_i for $i = 1, \ldots, n$. Suppose an LP deposits (resp. withdraws) the same amount of liquidity ΔL on range R_i, such that $L_i + \Delta L \geq 0 \, \forall i \in \{1, \ldots, n\}$, then he will deposit (resp. withdraw) the following amount of tokens into the pool:

$$\Delta x_r = \Delta L \cdot (\frac{1}{\pi_0^R} - \frac{1}{\pi_u}) \quad \text{and} \quad \Delta y_r = \Delta L \cdot (\pi_0^R - \pi_l) \quad \text{s.t.} \quad \pi^R = \begin{cases} \pi_u & \text{if } \pi_u < \pi \\ \pi_l & \text{if } \pi < \pi_l \\ \pi & \text{otherwise} \end{cases} \tag{1}$$

If an LP wishes to withdraw (burn) liquidity ΔL on price range R, then he would retrieve Δx_r and Δy_r real reserve tokens as specified above in (1), and in addition, earn $\Delta x_{\text{fee}} = \Delta L \cdot (\Phi_R^{\mathcal{X}}(t) - \Phi_R^{\mathcal{X}}(t_0))$ and $\Delta y_{\text{fee}} = \Delta L \cdot (\Phi_R^{\mathcal{Y}}(t) - \Phi_R^{\mathcal{Y}}(t_0))$, where accumulators $\Phi_R^{\mathcal{X}}$ and $\Phi_R^{\mathcal{Y}}$ are updated every time a trader swaps tokens while the square-root price is in range R. Suppose a_i tokens $\mathcal{X}$ are transfered into the pool in exchange for tokens $\mathcal{Y}$ via a swap operation, then $\Phi_R^{\mathcal{X}} \leftarrow \Phi_R^{\mathcal{X}} + \frac{a_i}{1-\phi} \cdot \frac{\phi}{L}$, where L is the available liquidity on range R. $\frac{a_i}{1-\phi}$ represents the input tokens actually used in the swap, which implies that $\frac{a_i \phi}{1-\phi}$ represents the accumulated fees from that operation and that $\frac{a_i}{1-\phi} \cdot \frac{\phi}{L}$ represents the amount of accumulated fees per unit of liquidity on the range R.

Impermanent Loss

A liquidity provider may provide liquidity and simultaneously hold tokens outside of the pool. If they withdraws their tokens at a time where the pool price is significantly different from the one at deposit time, the value of their tokens net of fees will be lower than the value of their original tokens

if they had not been deposited into the pool. This loss is only materialized when LPs withdraw their tokens from the pool, which is called *Impermanent Loss* (IL) [1].

Consider two strategies: (i) a liquidity providing stategy where tokens are deposited into a pool, and (ii) a strategy where tokens are withheld from the pool under. Assume $\mathbf{H_0}$: Tokens $\mathcal{X}$ and $\mathcal{Y}$ outside the pool can be valued using pool price. The value of strategies (i) and (ii) in reference numéraire token $\mathcal{Y}$ is denoted V_P and V_H, respectively.Suppose that an amount of liquidity ΔL, corresponding to token quantities Δx_r and Δy_r through equation (1), is added to unitary square-root price range $R := [\pi_l, \pi_u)$ at time t_0. Let $p_i = \pi_i^2$ denote the price at time t_i, then at a later time t_1,

$$V_P(t_1) = \Delta L \cdot \Big(\Big(\frac{1}{\pi_1^R} - \frac{1}{\pi_u} \Big) \cdot \pi_1^2 + (\pi_1^R - \pi_l) \Big) \tag{2}$$

Now suppose that quantities Δx_r and Δy_r are held outside of the pool (and under $\mathbf{H_0}$ are priced at $p_0 = \pi_0^2$ at time t_0).

$$V_H(t_1) = \Delta L \cdot \Big(\Big(\frac{1}{\pi_0^R} - \frac{1}{\pi_u} \Big) \cdot \pi_1^2 + (\pi_0^R - \pi_l) \Big) \tag{3}$$

The impermanent loss (which would be realized if the LP were to burn his tokens) is given by

$$V_P(t_1) - V_H(t_1) = -\Delta L \cdot (\pi_0^R - \pi_1^R) \cdot \Big(1 - \frac{\pi_1^2}{\pi_0^R \cdot \pi_1^R} \Big) \tag{4}$$

These results above can be generalized to a full liquidity curve. Let $\Delta L_\pi := \Delta L_i$ for $\pi \in [\pi_{l_i}, \pi_{u_i})$, then the function $\pi \mapsto \Delta L_\pi$ denote a liquidity curve. Since this mapping is piecewise constant and right-continuous with a left limit it allows for integrability w.r.t π.

Consider a liquidity provider adding a liquidity curve $(\Delta L_\pi)_\pi$ to the pool while keeping x_0 tokens $\mathcal{X}$ and y_0 tokens $\mathcal{Y}$ outside the pool (under $\mathbf{H_0}$). Assume the integrability condition

$$\int_1^\infty \frac{\Delta L_\pi}{\pi^2}\, d\pi + \int_0^1 \Delta L_\pi \, d\pi < \infty \tag{5}$$

At time t_0, when the price in the pool is p_0, their tokens admit the following $\mathcal{Y}$-value:

$$V_P(t_0) = x_0 \cdot p_0 + y_0 + p_0 \cdot \int_1^{+\infty} \frac{\Delta L_\pi}{\pi^2}\, d\pi + \int_0^{\pi_0} \Delta L_\pi \, d\pi < \infty \tag{6}$$

At time t_1, when the price in the pool is p_1, their tokens $\mathcal{Y}$-value net of swap fees is:

$$V_P(t_1) = x_0 \cdot p_1 + y_0 + p_1 \cdot \int_1^{+\infty} \frac{\Delta L_\pi}{\pi^2}\, d\pi + \int_0^{\pi_0} \Delta L_\pi \, d\pi < \infty \tag{7}$$

This representation lets us calculate the so-called "Greeks": (Delta) $\frac{\partial V_P(t_0)}{\partial p_0}$ and (Gamma) $\frac{\partial V_P(t_0^\pm)}{\partial p_0}$ which gives the sensitivity of the strategy's global position $V_P(t_0)$ with respect to the $X - Y$ exchange rate.

Fees Modeling

Token swaps are assumed to occur because of arbitrageurs between the pool and another reference exchange (due to pricing discrepancies see fig.fig. 2). Let latent price process $(p_t)_t$ be defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{0 \leq t \leq T}, \mathbb{P})$. Let $\mathbf{H_p}$: the latent price process follows an Itô dynamics of the form: $\mathrm{d}p_t/p_t = \mu_t \mathrm{d}t + \sigma_t \mathrm{d}W_t$ with progressively measurable drift $(\mu_t)_t$ and volatility $(\sigma_t)_t$: both μ and σ are bounded. This hypothesis allows for the creation of a risk-neutral probability measure $\mathbb{P}^*$ via the Girsanov transform.

Hence, swap trades are assumed to occur at discrete swap times when $(p_t)_t$ changes by $\gamma\%$ where $\gamma \in \{0.01, 0.03, 0.05\}$ is the swap fee paid out to liquidity providers. The price process p evolves at these discrete swap times and is assumed to coincide with the latent price process at these discrete times.

Consider an LP depositing a liquidity curve $(\Delta L_\pi)_\pi$ at time t_0. The amount of fees in $\mathcal{X}$ and $\mathcal{Y}$ accumulated over the period $[0, T]$ is given by:

$$\mathcal{F}((\Delta L_\pi)_\pi, (\sigma_t)_t) = \int_0^{+\infty} \Delta L_{\sqrt{b}} \frac{\mathbb{E}^*\left[A_T^b(p)\right]}{2 \cdot b^{3/2}} \, \mathrm{d}b \tag{8}$$

where A_T^b is the local time of the price process p at level a i.e. how much time p_t spends in price range $[b, b+\epsilon)$ over time range $[0, T]$. Exact fee calculations for a simulated price process, show an increase of fees in both tokens coincide with high local times (see fig)

Conclusion & Further Research

In this paper, M.Echenim et. al. sought to precisely define the way in which the Uniswap v3 protocol behaves to create an unambiguous knowledge-base for researchers and practitioners. However, since a lot of the paper spent time developing theory from the protocol open-source code itself, it leaves several investigations with real data open, namely, comparing fees predicted by equation (8) with those actually observed in the pool or performing a qunatitative study of arbitrage opportunities between calls/puts and Uniswap v3 pools. One of the challenges of using their equation to estimate fees is the inverse problem of estimating the drift and diffusion parameters of the price Itô price process p_t. Nevertheless, a qualitative analysis of fees reveals patterns so-far left unaddressed. Using The Graph protocol, I was able to query live statistics from a one of the most liquid Uniswap v3 pools corresponding to WETH (token $\mathcal{Y}$) USDC (token $\mathcal{X}$) exchanges for the time period Jan. 28th 2026 to T =Mar. 5th 2026. A qualitative analysis of fees shows a higher concentration of liquidity above p_T, and lower fees per unit liquidity below p_T, which could be indicative of a bearish sentiment as most LPs are choosing price ranges above the current price to add liquidity. This variability in the distribution could be incorporated in the model to model risk before a liquidity provider decides to mint or burn tokens.

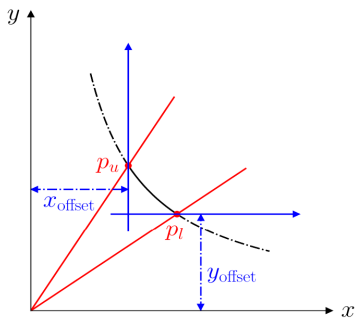

Figure 1: Representation of the virtual numbers (x, y) and offset numbers $(x_{\text{offset}}, y_{\text{offset}})$ of tokens

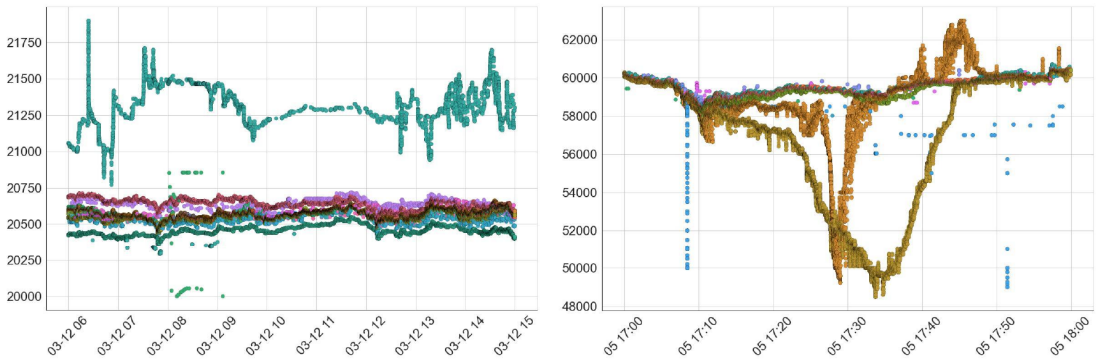

Figure 2: Centralized Exchange (CEX) prices differ from one another with each color represents a differet exchange. The trade prices are for the BTC-USD exchange rate on March 12th 2023 (left) and BTC-EUR exchange rate on March 5th 2024 (right))

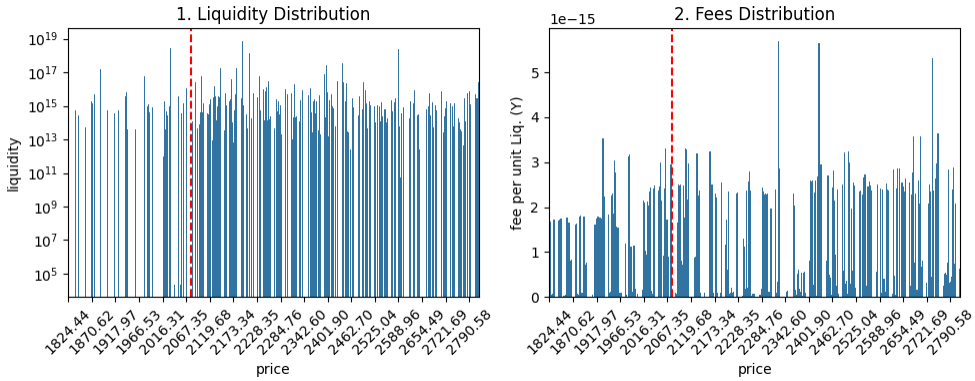

Figure 3: Liquidity distribution and distribution of fees per unit-liquidity L for WETH (token $\mathcal{Y}$) USDC (token $\mathcal{X}$) for time period Jan. 28th, 2026 to Mar. 5th 2026. Price on Mar. 5th 2026 indicated by red-dashed line

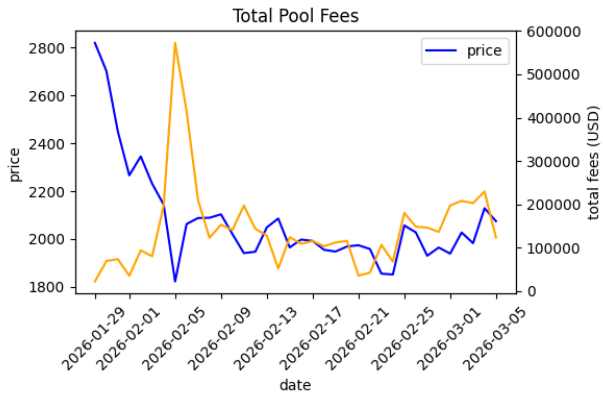

Figure 4: Total liquidity pool fees accumulation from swap transactions between Jan. 28th, 2026 and Mar. 5th 2026

References

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